

Compare the turtle behavior with noise off to noise on for several values of q . How does the noise affect the movement?

When $q = 0$, the noise has relatively little effect because the butterflies are moving randomly regardless of elevation. At $q = 0.4$, the difference is much more noticeable. Without noise, the butterflies follow highly consistent, artificial-looking paths toward the same areas, while adding noise produces much more varied movement and eliminates the obvious tunnel-like paths. At $q = 1$ without noise, the butterflies follow a nearly straight path toward the highest hill. With noise, however, they can stop at smaller local peaks instead of reaching the highest hill.

Does this give you any insight into why the paths in the original (noise-free) model look artificial and unlike what you might expect butterflies to do in the real world?

This happens because uphill only considers the elevations of neighboring patches. It does not know where the highest point in the entire landscape is. Adding noise creates many small local maxima, so a butterfly can reach one and have no higher neighboring patch to move to. At $q = 0.9$, the occasional random movement allows butterflies to escape some of these local maxima and continue moving toward higher areas. This also helps explain why the original model looks artificial. The noise-free sine-wave landscape is extremely regular, so the same movement rules repeatedly produce the same paths. Real landscapes have much more small-scale variation, which would make movement less predictable.